



50 MW (AC) SOLAR PARK AT SUTIAKHALI, GOURIPUR, MYMENSINGH



Document Prepared by Shenzhen CTI International Certification Co., Ltd

Project Title	50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh
Report Title	Joint Validation & Verification Report of 50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh
Version	2.0
Report ID	CTI/NB-2022-0801
Verification Period	04-Nov-2020 to 31-Aug-2022
Client	China International Engineering Consulting Corporation (Hong Kong) Limited
Pages	59
Date of Issue	29-Oct-2022
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Summary:

Shenzhen CTI International Certification Co., Ltd (hereafter referred to as CTI) has been commissioned by China International Engineering Consulting Corporation (Hong Kong) Limited to perform the joint validation and verification of “50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh” in Bangladesh (VCS Project ID: 3545) (hereafter referred to as “the project activity”) and reported in the joint Validation & Verification Report.

The validation and verification include confirming the project’s description, baseline, additionality, monitoring plan to be compliance with relevant VCS and host party criteria, and implementation of the monitoring plan in the Joint Project Description & Monitoring Report (hereafter referred to as “Joint PD & MR”) to the application of the CDM methodology ACM0002 (Version 20.0): “Grid-connected electricity generation from renewable sources”.

The purpose of this project activity is to generate clean form of electricity through renewable solar energy source for sale of electricity to the grid. The project activity involves the generation of electricity through a greenfield solar photovoltaic plant with a capacity of 50 MW (AC) at a 170-acre plot of land approximately 7km South from the town center of Mymensingh City, Bangladesh. The solar plant supplies the generated electricity to the Bangladesh Power Development Board (BPDB), which is owned and operated by Power Grid Company of Bangladesh Ltd. (PGCB). The project aims to reduce the dependency on fossil fuel power plants for electricity by substituting carbon intensive energy with solar energy. In the baseline scenario, the electricity delivered to the grid by the project activity would have otherwise been generated by operation of fossil fuel grid-connected power plants and by addition of new generation sources. The project activity is estimated to result in emission reductions with 55,302 tCO_{2e} per year and 387,113 tCO_{2e} in total during the first crediting period of 7 years, 04/11/2020 - 03/11/2027.

In the course of the validation and verification, 6 Clarification Requests (CLs) and 4 Corrective Action Requests (CARs) were raised and successfully closed. No Forwarded Action Request (FAR) was raised.

Validation Summary

The review of the project design documentation, monitoring report and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and stakeholders have provided CTI with sufficient evidence to validate the fulfillment of the stated criteria.

The purpose of the validation is to have a thorough and independent assessment of the proposed project activity against the applicable VCS requirements, in particular, the project’s description, baseline, additionality, monitoring plan to be compliance with relevant VCS and host party criteria. These are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of emission reductions. CTI’s objective is to perform a thorough, independent assessment

of the validation of the project activity. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of Verified Carbon Units (VCUs). Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to Verra registry.

The validation scope is defined as an independent and objective review of the Joint PD & MR. The Joint PD & MR is reviewed against the relevant criteria and guidance documents provided by VCS which included: VCS Program Guide (Version 4.2), VCS Standard (Version 4.3), Program Definitions (Version 4.2), Registration & Issuance Process (Version 4.2) and in line with the VCS Validation and Verification Manual applicable at the time in order to confirm that the project meets the applicability conditions of the selected baseline and monitoring methodology ACM0002 (Version 20.0) and also assess the claims and assumptions made in the Joint PD & MR without limitation on the information provided by the project proponents.

Verification Summary

CTI has performed the verification of the emission reductions reported for the project activity "50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh" (VCS Project ID: 3545) for the monitoring period 04/11/2020 to 31/08/2022 to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as VCUs under the VCS Standard (Version 4.3).

The verification performed on the basis of VCS Program Guide version 4.2 and VCS standard version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report. Hence, CTI is able to certify that the emission reductions from the "50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh" during the period amount to 99,987 tCO₂e.

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1 INTRODUCTION

1.1 Objective

Shenzhen CTI International Certification Co., Ltd (hereafter referred to as CTI) has been commissioned by China International Engineering Consulting Corporation (Hong Kong) Limited to perform a joint validation and verification of "50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh" in Bangladesh (VCS Project ID: 3545) (hereafter referred to as "the project activity").

The objective of validation is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCU. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to Verra registry.

The objective of verification is the periodic independent review and ex-post determination by an accredited Third Party of the monitored reductions in GHG emissions that have occurred as a result of the VCS project activity during a defined verification period. A verification statement is the written assurance that, during a specific period in time, a project activity achieved the emission reductions as verified. For the project activity, the verification was to verify and provide a verification statement of emission reductions reported for the period 04/11/2020 to 31/08/2022.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the joint Project Design and Monitoring Report to validate that (a) the project design is actual, (b) the baseline scenario is correctly defined as per the applied methodology and related tools, (c) the project is additional, (d) the monitoring plan can be implemented and is transparent and adequate and (e) all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature. The joint PD & MR is reviewed against the criteria stated in VCS standard (version 4.3) /36/ and the approved baseline and monitoring methodology ACM0002 (Version 20.0). The validation was based on VCS standards (version 4) /36//37/ applying auditing techniques. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The scope of the verification is to verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan; evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement; verify that reported GHG emissions data is sufficiently supported by evidence. The verification is conducted as per VCS Program Guide (version 4.2), VCS Standard (version 4.3), other relevant requirements defined by Verra and the approved methodology ACM0002 (Version 20.0) applied by the project. The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The project activity is expected to generate average of 55,302 VCUs per year, and therefore categorized as a project as per the VCS rules, in regards to its scale, subject to a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

The assessment was conducted to provide a reasonable level of assurance of compliance against the defined audit criteria and materiality threshold within the audit scope. Based on the audit findings, the validation statement reasonably assures that the GHG assertion is materially correct and is a fair representation of the GHG data and information.

1.4 Summary Description of the Project

The project activity is implemented by HDFC SinPower Limited, which aims to use of solar power to generate renewable and clean electricity delivered to Bangladesh Power Development Board (BPDB), which is owned and operated by Power Grid Company of Bangladesh Ltd. (PGCB). It locates at a 170-acre plot of land approximately 7km South from the town center of Mymensingh City, Bangladesh and geographical coordinates of the project activity are 24°42'13.39"N (24.70372°) and 90°27'49.81"E (90.46384°). The total installed capacity of the project activity is 50MW at AC level (60.177 MWp at DC level) with lifetime of 20 years and the expected average annual net electricity generation during the whole operation lifetime is 92,192 MWh.

The project activity achieves greenhouse gases (GHGs) emission reductions by avoiding CO₂ emissions from the business-as-usual scenario electricity generation of those fossil fuel-fired power plants connected to the PGCB, generates social and environment benefits and contributes to sustainable development of the local community, the host country and the world, which has been verified by on-site inspection, and checking the related documentations, interview with the project implementer. The project is expected to achieve GHG emission reductions of 55,302 tCO₂e annually and 387,113 tCO₂e in total in the first crediting period.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

A project specific validation and verification plan was developed to guide the validation and verification auditing process to ensure efficiency and effectiveness.

The purpose of the validation and verification is to present a risk assessment for determining the nature and extent of validation and verification procedures necessary to ensure the risk of auditing error is reduced to a reasonable level. According to the ISO14064-3, the criteria are the policy, procedure or requirement used as reference against which evidence is compared. The validation and verification were performed through means of the following three phases in accordance with the requirement of the applied methodology, and VCS Standard (version 4.3) and other relevant VCS requirements.

- A desk review of the joint PD & MR and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the joint validation and verification report as well as statement.

CTI completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of (if applicable):

- Project Details;
- Application of Methodology;
- Estimated GHG Emission Reduction and Removals;
- Monitoring;
- Safeguards etc.

CTI also verified the emission reductions through assessing all factors and issues that constitute the basis for emission reductions from the project, including:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

CTI assessed that the reported information in the joint PD & MR are complete and accurate in question. This joint validation and verification report describes the findings of this assessment. The information of the assessment team has been included in below of this report.

2.2 Document Review

The joint PD & MR /1/ and supporting background documents related to the project implementation were reviewed. Documents review was conducted to ensure consistency with and identify any deviation from VCS program requirements. Desk review included an examination of the project design details, baseline scenario, additionality, ex ante and monitoring data and parameters, and ex ante quantification of GHG emission reductions. The assessment team also used additional documentation by third parties like host party legislation, technical reports referring to the project design details, baseline scenario, additionality, monitoring or to the basic conditions and technical data.

Furthermore, the joint PD & MR /1/ and emission reduction calculation spreadsheet of monitoring period from 04/11/2020 to 31/08/2022, relevant decisions, clarifications and guidance from the Verra, other information and references relevant to the project activity have also been assessed.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided, including a review of the data and information presented to verify their completeness; a review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and an evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

The references used in the course of this validation and verification are summarized in Appendix 2. The validation and verification were performed basing on the documents check and site inspection/measurements, refer to the Section 3 of this report for the validation process detail, Section 4 of this report for the verification process detail.

According to the sectoral scopes / technical area and experiences in the sectoral or national business environment, CTI has composed a project assessment team in accordance with the appointment rules in CTI. The composition of CTI assessment team has to be approved by CTI ensuring that the required skills are covered by the team. The four qualification levels for team members that are assigned by formal appointment rules as below:

Function	Name	Technical competence	Task Performance*
Team Leader	Lin Wu	1.1, 1.2, 2.1, 3.1, 4.1, 5.1, 5.2, 10.1, 11.1, 11.2, 12.1, 13.1, 13.2	☒ DR ☒ SV ☒ RP ☐ TR
Team member	Zhang Wenting	1.2	☒ DR ☒ SV ☒ RP ☐ TR
Local expert	Nasif Shams	1.1, 1.2	☒ DR ☒ SV ☒ RP ☐ TR

Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒ DR ☐ SV ☐ RP ☒ TR
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*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

2.3 Interviews

CTI was commissioned by the project proponent (PP) to perform a joint validation and verification on the project activity in July 2022. The project started operation on 04/11/2020, which was considered as the project start date. The project activity shall be completed the validation within 2 years of project start date following the VCS rules. Due to the spread of the Coronavirus (COVID-19) and China's aviation regulation, CTI cannot perform the on-site visit to conduct validation and verification on the project activity within the timeline. Thus, PP submitted exemptions request for a site visit to the Verra via email on 30/09/2022 and received the response from the Verra on 01/10/2022. It mentioned VCS rules do not specifically require onsite audits, but all audits must be able to demonstrate a reasonable level of assurance and therefore audits conducted remotely may face additional scrutiny in how conformance was assessed remotely.

To guarantee the authenticity, specialization and validity of the validation and verification on the project activity, CTI's validation and verification team has included a local expert when CTI assigned the assessment team during the contract review. Further, in response to the effect and limitation from COVID-19 on the audits, CTI has prepared the procedure about remote auditing to ensure the reasonable level of assurance and conformance with the relevant requirement of VCS rules. Hence, the remote validation and verification process was conducted by CTI's assessment team on 24/10/2022, and all assessment team co-participated in the online review. Several kinds of auditing skill were conducted by the assessment team to perform the remote validation and verification. The interview with project manager, project operators, and local stakeholders was conducted by Tencent Conference. The remote on-site inspection was conducted by the mobile video to check the solar PV module, boost inverter system, DC system, substation, monitoring metering system, power connection diagram, etc. Hence, CTI considered the remote validation and verification ensure the reasonable level of assurance and conformance with the VCS rules and CTI's procedures. The following is a list of the main interviewees and subject.

Interviewed personnel	Role	Organization	Subject
Chen Wei	Managing Director/Project Manager	HDFC SinPower Limited	- Status of the project; - Applicability of selected methodology; - Baseline assessment;

Interviewed personnel	Role	Organization	Subject
Wang Hongqiang	Site Manager		- Additionality assessment; - Emission reduction calculations; - Monitoring plan; - Stakeholder consultation process and its outcomes; - Operation of the project activity; - Implementation of the monitor plan of the project activity; - Data collection and data achievement; - Calibration of meters and equipment maintenance; - Data collection and ER calculation.
Zhang Jiafu	Operations and Maintenance Manager		
Kamal Khan	Local resident	Mymensingh City	- The process and participation of the stakeholder consultation; - The impact of the project activity; - The implementation of the mitigation measures.
Manik Rahman	Local resident		
Mihir Lal	Officer	Mymensingh division	
Zhang Hui	Project Leaders	China International Engineering Consulting Corporation (Hong Kong) Limited	- Data collection and ER calculation.
Li Xin	Project Manager		
Deng Yuheng	Project Manager		
Wang Fei	Finance Specialist		

2.4 Site Inspections

The assessment team performed the remote audit on 24/10/2022. The interviewed personnel and objective are listed in above table.

2.5 Resolution of Findings

As an outcome of the validation process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;

- b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the project participants;
- c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

All CARs and CLs raised during validation shall be resolved prior to submitting a request for issuance.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for CTI's positive conclusion on the joint PD & MR /1/. 6 Clarification Requests (CLs) and 4 Corrective Action Requests (CARs) were raised and successfully closed. No Forwarded Action Request (FAR) was raised by CTI. And communications were made between the PP and CTI to guarantee the transparency of the validation and verification process, the concerns raised, and responses given are summarized below in the Appendix 4.

The final joint PD & MR (Version 02 dated 27/10/2022) serves as the basis for the final assessment presented. Additional changes to the project during the validation and verification process are not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
Project Description and monitoring report	0	1	0
Description of project activity	3	1	0
Application of selected baseline and monitoring methodology and selected standardized baseline	---		
- Applicability of methodology and standardized baseline	0	0	0
- Deviation from methodology	0	0	0
- Clarification on applicability of methodology, tool and/or standardized baseline	0	0	0
- Demonstration of additionality	3	0	0
- Emission reductions	0	1	0
- Monitoring plan	0	0	0
- Stakeholders consultation process	0	0	0
- SDG	0	1	0
- Public comments	0	0	0
Others (please specify) - Matter related to double counting - for validation	0	0	0
Others (please specify) - Matter related to Emission reduction calculation - for verification	0	0	0
ER achieved, Matter related to feeder details, breakdown and Calibration - for verification	0	0	0
Total	6	4	0

2.5.1 Forward Action Requests

None FAR was raised during the validation and verification process.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project

- *Type*

By interviewing with PP and documents review, it is verified by CTI the project activity is a newly-built grid-connected solar power project, which is to use solar power resource for electricity generation. Therefore, it is confirmed that the sectoral scope of the project activity falls into Sectoral Scope 01: Energy industries (renewable - / non-renewable sources) as per the UNFCCC Standard "Applicability of sectoral scopes" (Version 01.0) /45/.

- *Technologies and measures implemented*

The project activity involves the generation of electricity through a greenfield solar photovoltaic plant with a capacity of 50 MW (AC) at a 170-acre plot of land approximately 7km South from the town center of Mymensingh City, Bangladesh. The solar plant supplies the generated electricity to the Bangladesh Power Development Board (BPDB), which is owned and operated by Power Grid Company of Bangladesh Ltd. (PGCB). The project aims to reduce the dependency on fossil fuel powered power plants for electricity by substituting carbon intensive energy with solar energy. In the baseline scenario, electricity delivered to the grid by the project activity would have otherwise been generated by operation of fossil fuel grid-connected power plants and by addition of new generation sources. In the FSR, it stated the project activity will install 139,947 photovoltaic modules with unit output of 430 W, and the total capacity is 60.177 MWp at DC level and 50 MW at AC level. However, by checking the equipment purchase contract and nameplate of photovoltaic modules, the assessment team found the photovoltaic modules included both 50,736 photovoltaic modules with unit output of 430 W and 88,356 photovoltaic modules with unit output of 435 W. The actual total capacity of the project activity was 60.251 MWp at DC level and 50.05MW at AC level, which was slightly higher than the design value. PP explained the change was due to the rapidly technical development of photovoltaic modules and limitation of photovoltaic equipment market. Based on the sectoral experience, the assessment team considered the change above is the common issues in the solar power field, and the slight change in the installation capacity will not affect greatly the power generation and financial values. Hence, CTI confirmed the change is reasonable to reflect the operation practice.

CTI has further verified the project implementation by interview and checking the operation and maintenance records, the power purchase agreement (PPA), as well the power connection diagram. It

was confirmed that the project started construction on 22/09/2019 and all solar arrays put into full operation since 04/11/2020 supplied to the PGCB. By replacing the equivalent electricity generated by fossil fuel fired power plants from the PGCB, it's estimated that the project could achieve average GHG emission reductions of 55,302 tCO₂e annually and 387,113 tCO₂e in total in the first crediting period.

▪ *Eligibility of the project activity*

For the validation, the joint PD & MR has described and provided the justification on the project's eligibility under the scope of the VCS Program as per VCS Standard (Version 4.3). The assessment is provided as below:

The project activity generates GHG emission reductions including CO₂ which belong to the six Kyoto Protocol greenhouse gases attributed to Scope 1 of Section 2.1.1 of VCS Standards (Version 4.3).

The project activity adopted CDM methodology ACM0002 (version 20.0), which is approved under a VCS approved GHG program, attributed to Scope 4 of section 2.1.1 of VCS Standards (Version 4.3).

Via checking the joint PD & MR of the project, it is verified that the project is implemented to utilize solar power for electricity generation and deliver to PGCB. It reduces GHG emissions by the fossil-fuel based power plant of PGCB, and the additionality is verified as actual, thus there is no assumption of having generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction for this project.

In conclusion, as a VCS project, the project remains eligible under the scopes of the VCS Program.

Project design, including eligibility criteria for grouped projects

The project has not been designed to include a single installation of an activity, multiple project activity instances, or as a grouped project.

Project Proponent and other entities involved in the project

By checking the Business License /12/, approvals of EIA and FSR /6//7/, and via online inspection and interview, the assessment team confirmed the project is owned and operated by HDFC SinPower Limited. Therefore, HDFC SinPower Limited is confirmed to be the project proponent of the project.

The other entity involved in the Project is China International Engineering Consulting Corporation (Hong Kong) Limited, which is appointed by the project proponent as the consultancy for VCS development as verified by interviewing with the representative from the project proponent and the consultancy company. The contact information reported in section 1.6 of the joint PD & MR has been confirmed to be correct through checking the Business License /13/ and interviewing the representative of the company.

Ownership

By checking the Business License /12/, approval of EIA and FSR /6//7/, Purchase Contract of the equipment /14/ and Power Purchase Agreement /10/, the assessment team confirmed that the project

proponent HDFC SinPower Limited has the legal right to control and operate the project activity and has full legal and beneficial title to all of the GHG reductions generated by the Project.

Project Start Date

The project start date is 04/11/2020 on which the project starts operation. By checking the notification of grid connection/11/ and interviewing with PP, it is verified the project start date is correct and in line with the VCS standard (Version 4.3).

Project Crediting Period

CTI confirms the crediting period for the project activity is as below:

Crediting Period Start date: 04/11/2020

Crediting Period End date: 03/11/2027

This project adopts twice renewable for a total of 21 years.

Project Scale and Estimated GHG Emission Reductions or Removals

As the estimated annual average GHG emission reductions are 55,302 tCO₂e which is lower than 300,000 tCO₂e, the project is classified as “Project” under VCS standard. The total estimated GHG emission reductions during the first VCS crediting period (04/11/2020 - 03/11/2027) are 387,113 tCO₂e.

By checking emission reductions calculation spreadsheet /2/, the assessment team confirmed that the estimated GHG emission reductions of the project activity are as follows:

Year	Estimated net GHG emission reductions or removals (tCO ₂ e)
04/11/2020~03/11/2021	56,076
04/11/2021~03/11/2022	55,818
04/11/2022~03/11/2023	55,560
04/11/2023~03/11/2024	55,302
04/11/2024~03/11/2025	55,044
04/11/2025~03/11/2026	54,786
04/11/2026~03/11/2027	54,527
Total	387,113
Total number of crediting years	7 years
Average annual ERs	55,302

Project location

The project located at a 170-acre plot of land approximately 7km South from the town center of Mymensingh City, Bangladesh. By check the map against GPS location shared by the site staff during the remote online inspection, it is confirmed that the central geographical coordinates of the proposed project are 24° 42' 13.39"N (24.70372°) and 90° 27' 49.81"E (90.46384°).

Conditions prior to project initiation

The assessment team during the desk review and remote online inspection confirms that the project is a grid-connected Solar power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. The baseline as described in section 3.4.4 of this report will continue to be the baseline in the absence of project activity.

Project compliance with applicable laws, statutes and other regulatory frameworks

By checking the approval for EIA and FSR as well as relative laws and regulation, i.e. Renewable Energy Policy of Bangladesh/22/, it is confirmed that the project activity is in complicate with all laws and regulations in Bangladesh.

Participation under other GHG programs

The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. A declaration from PP has been provided and verified by CTI. Further, the assessment team checked the registries of CDM EB and Gold-standard respectively, through website <http://cdm.unfccc.int/>, <http://www.goldstandard.org/>.

The Project is not rejected by other GHG programs. A declaration for the same has been provided and verified by CTI. Further, the assessment team checked the registries of CDM EB and Gold-standard respectively, through website <http://cdm.unfccc.int/>, <http://www.goldstandard.org/>.

Other forms of credit

The assessment team confirms that the net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. Firstly, the assessment team confirmed there was not emission trading programs in Bangladesh by interviewing with local stakeholders. Further, the assessment team checked the REC Mechanism database found that the project activity is not registered under REC mechanism. By checking the statement of no-double counting by PP/17/, it is confirmed that the project activity has not received nor sought any other forms of credit. Thus, the assessment team concluded that the project activity not involved on other Emissions trading programs and other binding limits.

The assessment team confirmed the Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

Sustainable development contributions

Via the remote site inspection and by checking the evidences provided, CTI verified that the project contributes the sustainable development through the following aspects:

- SDG13 “Take urgent action to combat climate change and its impacts”:

GHG emission reductions will be generated through the implementation of the project activity by avoiding the consumption of the equivalent raw materials and reducing related energy consumption.

- SDG7 “Ensure access to affordable, reliable, sustainable and modern energy”:

The project reduces the consumption of the fossil fuels of BPDB, which contributes to increasing the share of renewable energy and improve Bangladesh’s energy security.

- SDG8 “Promote inclusive and sustainable economic growth, employment and decent work for all”:

The implementation of the project will offer job opportunities for local people during both the construction and operational period.”

Additional information relevant to the project

▪ *Leakage management for AFOLU projects*

The project activity is not a AFOLU project, therefore, this section is Not applicable.

▪ *Commercially sensitive information*

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to validation team for analysis which lead to positive conclusion for this validation.

Overall, it is confirmed that the joint PD & MR is accurate, complete, and provides an understanding of the nature of the project.

3.2 Participation under Other GHG Programs

The project activity is seeking registration under VCS programme only. The start date of the project activity is 04/11/2020, which is the operation date of the project activity. Hence it can be confirmed that the project is complying with all the applicable VCS rules, including project start date and project crediting period requirements.

During the validation process, the assessment team has reviewed the declaration/17/ submitted by PP to confirm that GHG emission reduction credits from the project activity have not been registered under other GHG program. It can be concluded that the project is eligible to participate under the VCS Program.

3.3 Safeguards

3.3.1 No Net Harm

The assessment team, by checking the environmental impact assessment (EIA)/6/, confirmed that the environmental impact of the project is exceptionally low, i.e., no negative impact, and that the project will also generate social and economic benefits through the development of employment, local business development and community skills.

In addition, the project activity will help reduce greenhouse gas emissions by replacing the same amount of electricity generated by fossil fuel power plants.

3.3.2 Local Stakeholder Consultation

As required by the VCS, it was necessary to invite relevant stakeholders prior to the validation process. The relevance of the dates was checked by the assessment team during the validation site visit.

A local stakeholder consultation was conducted during the EIA with the aim of gathering the views of local residents and project owner employees. As part of the consultation process, project representatives explained to all stakeholders the type of development and the technology used to generate electricity to increase stakeholder awareness of the project. Stakeholders were also informed about the potential employment opportunities and other benefits that could be generated through the project and the importance of establishing a solar PV power plant. A questionnaire /33/ was distributed to participants including local residents, the government EPA and other relevant stakeholders in 03/2017. The questionnaires were reasonably designed to assess the impact of the project on the local environment and social economic development. 30 out of 30 questionnaires were returned with a return rate of 100%. By checking the questionnaires, the evaluation team confirmed that there were no negative comments from the local stakeholders on the project construction activities.

In addition, through remote online interviews, the assessment team confirmed that there was regular communication with stakeholders during project implementation. Major implementation schedules or changes to the project have been and will be communicated to the local government, neighborhood councils, and local residents. Local government agencies and authorities will conduct random checks on the implementation of the project and make recommendations to the project owner on the rectification issues involved. By checking the meeting minutes /34/, the assessment team confirmed the project owner conducted focus group discussions on 26/12/2020 and 20/11/2021 to collect relevant comments and suggestions. No negative comments were received on the project during this monitoring period.

Therefore, the assessment team was able to confirm that the stakeholder meetings were adequate and appropriate.

3.3.3 Environmental Impact

By reviewing the EIA/6/ and online interviews, it was confirmed that the project would not have a negative impact on the environment.

3.3.4 Public Comments

The assessment team noted that the project activity was open for public comment from 11/08/2022 to 10/09/2022. The detail was checked by the assessment team in the following web platform: <https://registry.terra.org/app/projectDetail/VCS/3545>.

During the public period, no comments were received.

3.3.5 AFOLU-Specific Safeguards

The project activity is not an AFOLU project. For non-AFOLU projects, this section is not required.

3.4 Application of Methodology

3.4.1 Title and Reference

The assessment team checked following methodology and tools to be applicable for the project activity.

The details are as below:

Methodology:

ACM0002 “Grid-connected electricity generation from renewable sources” (Version 20.0)

Reference: <https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG>

The applied tools:

Tool 01- “Tool for the demonstration and assessment of additionality” - (Version 7.0.0)

Reference: <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

Tool 07 – “Tool to calculate the emission factor for an electricity system” -(Version 07.0)

Reference: <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

Tool 24 – “Common practice” -(Version 03.1)

Reference: <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

Tool 27- “Investment analysis”- (Version 11.0)

Reference: <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf>

3.4.2 Applicability

The criteria and assessment of ACM0002 (Version 20.0) are in the following table:

Criteria	Analysis	Validation Team’s opinion
This methodology is applicable to grid-connected renewable energy power	The project is a greenfield grid-connected renewable	Via remote site inspection and checking

Criteria	Analysis	Validation Team's opinion
generation project activities that: a) Install a Greenfield power plant; b) Involve a capacity addition to (an) existing plant(s); c) Involve a retrofit of (an) existing operating plants/units; d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s).).	power generation project.	the FSR, it is confirmed that the project is a grid-connected Greenfield solar power plant.
The methodology is applicable under the following conditions: (a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The project is a newly built solar power project and the project activity involves the installation of the solar power plant.	Via remote site inspection and checking the FSR, it is confirmed that the project is a Greenfield solar power plant.
In case of hydro power plants, one of the following conditions must apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs,	Not applicable, the project is not a hydro power plant, so this applicability condition does not need to be considered.	Via remote site inspection and checking the FSR, it is confirmed that the project is a solar power project, this is not applicable.

Criteria	Analysis	Validation Team's opinion
where the volume of the reservoir(s) is increased and the power density calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.		
In the case of integrated hydro power projects, project proponent shall: Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power	Not applicable, the project is not a hydro power plant, so this applicability condition does not need to be considered.	Via remote site inspection and checking the FSR, it is confirmed that the project is a solar power project, this is not applicable.

Criteria	Analysis	Validation Team's opinion
project; or Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.		
The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	The project does not involve switching from fossil-fuels to renewable energy sources at the site of the project activity and the project is not a biomass-fired power project.	Via remote site inspection and checking FSR, it is confirmed that the project is a Greenfield solar power plant, does not involved in switching from fossil fuel to renewable energy and biomass fired power plant.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	Not applicable, the project is a newly built solar power project.	Via remote site inspection and checking the FSR, it is confirmed that the project is a solar power project, this is not applicable.

According to the table above, the assessment team confirms the project meets all the applicable criteria of ACM0002 (Version 20.0).

For related tools, the assessment team confirms all applicable criteria have been met confirmed by remote site inspection and checking the FSR.

3.4.3 Project Boundary

Project boundary has been ascertained and confirmed by the assessment team according to ACM0002 (Version 20.0) - "The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to".

Via interviewing, checking the PPA /15/and Grid Access Agreement /16/, it is confirmed that the project boundary includes the project site and all power plants connected physically to BPDB.

The calculation of net electricity supplied to grid is under purview of state electricity board and project activity owner or project activity implementer does not have any control on it. Thus, for project activity, net electricity supplied to the grid is the monitoring parameter which is used for ER calculations.

The sources and GHG gases involved for proposed Project activity are as below.

Source		Gas	Included?	Justification/Explanation
Baseline	CO ₂ emissions from electricity generation in fossil fuel-fired power plants that is displaced due to the project activity.	CO ₂	Yes	Major emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
		Other	No	Minor emission source
Project	The solar power plant	CO ₂	No	The project is a solar power plant project. According to the methodology, it does not involve project emissions.
		CH ₄	No	The project is a solar power plant project. According to the methodology, it does not involve project emissions.
		N ₂ O	No	The project is a solar power plant project. According to the methodology, it does not involve project emissions.
		Other	No	Project activity does not emit other gas.

3.4.4 Baseline Scenario

The baseline scenario is in accordance with ACM0002 (Version 20.0) /43/ that in the absence of the proposed project activity, given that the project is additional, the same amount of electricity would have been generated by the operation of existing and new grid-connected power plants in the BPDB dominated by fossil fuel based power plants, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system"/44/.

The BPDB is dominated by fossil fuel-fired power plants. It is deemed likely that fossil fuel-fired power plants will continue to dominate the power sector due to the local availability of low-cost fossil fuel. It is expected that renewable capacity additions will not have significant effects on the mix of the BPDB during the first crediting period /2//30/.

Therefore, the baseline scenario is the continuation of the current situation, i.e., generation of equivalent electricity from the BPDB connected power plants.

All the assumption and data used by the project participants are listed in the PD & MR and/or supporting documents. All documentation relevant for establishing the baseline scenario are correctly quoted and interpreted in the PD & MR. Assumptions and data used in the identification of the baseline scenario are justified appropriately, supported by evidence and can be deemed reasonable. Relevant national and/or sectoral policies and circumstances are considered and listed in the PD & MR.

3.4.5 Additionality

The additionality of the project, as required by ACM0002 (Version 20.0), is demonstrated by Tool 01 “Tool for the demonstration and assessment of additionality” (Version 7.0.0).

During conceptualization of the project activity, board of directors of PPs considered the VCS revenue to improve the project financials. During the board meeting for board of Directors decided that they would consider VCS revenue for their project activity. In continuation to the board decision, PP issued the respective purchase order for the supply of solar plant.

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

The proposed project activity is not the first-of-its-kind. Hence not applicable.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity:

Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed VCS project activity.

The purpose of the project activity is to generate electrical power using solar energy and feed the electricity generated to the grid. Hence, the following alternatives are considered:

Alternative 1: The proposed project activity not undertaken as a VCS project activity.

The PP could proceed with the implementation of the project without Carbon credit benefits. The electricity produced from the renewable energy project would have been sold to the grid. This is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, the Project activity is not feasible without revenues from sale of Carbon Credits. This argument has been discussed in Step 2 of the Additionality section.

Alternative 2: No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid i.e. Continuation of the present situation. The PP would have continued without investment in Project activity with usual business activities. The grid would continue with the fossil fuel-based power projects and this would result in GHG emissions. Hence, the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1a: All the realistic alternatives for the project activity have been enlisted above.

Thus, though two alternatives are mentioned above as per step of additionality tool, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology.

Sub-step 1b: Consistency with mandatory laws and regulations:

The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g., to mitigate local air pollution. As verified in the results above, renewable energy generation is encouraged in Bangladesh, so both the alternatives identified above are found to comply with the mandatory laws and regulations.

Step 2: Investment analysis

Determine whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in Step 1, without the revenue from the sale of emission reductions credits. To conduct the investment analysis, use the following sub-steps:

Sub-step 2a: Determine appropriate analysis method

The Project activity envisages to export the power to the grid and the revenues from the sale would be generated in accordance with the terms and tariffs established in the PPA. Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity.

In the absence of the project activity grid electricity would have been the obvious choice for the Project which requires no investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option I and Option II, the use of Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates the evaluation of the Project by the PP before the decision to undertake the project was taken and management approval granted.

Choice of Financial Indicator:

According to the “Tool for demonstration and assessment of Additionality”, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR (EIRR).

Choice of Benchmark:

As per “Investment analysis” (Version 11.0), Required/expected returns on equity are appropriate benchmarks for an equity IRR. The equity IRR is considered as the financial indicator and the benchmarks used is cost of equity. Hence, the benchmarks used are applicable to the project activity and the type of IRR calculation presented.

At the time of decision made of project activity, methodological Tool 27 “Investment analysis” (Version 11.0) was the latest available tool to PP at the time of the decision making. Hence, PP has considered the same tool for default value of return on equity. The default value of Return on Equity for Group-1 projects in Bangladesh is 11.72% following Tool 27 (Version 11.0).

As per paragraph 16 of the above-mentioned document, “In situations where an investment analysis is carried out in nominal terms, project participants can convert the real term values provided in the table below to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used”.

Default Value Benchmark:

As per para 19 of Tool 27 (Version 11.0), the cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

Appendix in Tool 27 (Version 11.0) specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Bangladesh = 11.72%.

The Required return on equity (benchmark) was computed in the following manner /19/:

$$\text{Nominal Benchmark} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

- Default value for Real Benchmark = 11.72% (as per Appendix of Tool 27 (Version 11.0))

- Inflation rate forecast for by the Asian Development Bank of Bangladesh

Benchmark estimation:

For Group 1 project category, Appendix in Tool 27 (Version 11.0) specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Bangladesh = 11.72%.

CTI confirms that the data published by the Asian Development Bank/18/ on future inflation rates after the start of project activities are only available for the next four years and therefore takes the average forecast inflation rate of 5.8% for the next four years. The nominal benchmark is calculated to be 18.20% as per the above formula.

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

The PP has evaluated the Post tax Equity IRR for 20 years, which satisfies the requirement in paragraph 6 of Tool 27 (Version 11.0) that the investment analysis should be conducted for at least 10 years. And it is calculated based on the cash outflows from and cash inflows into the project activity.

CTI validation team has verified and crosschecked all the input values used for the EIRR calculations in the spreadsheet submitted for registration. The input parameters used in the financial analysis are sourced from the FSR /7/.

The approvals of the project (also the FSR) were issued by the government from Bangladesh /8/ and China /9/, respectively, which were only after it passes the assessment of the sector experts designated by the government. Therefore, the assessment team deemed that a FSR /7/ can be regarded as a reliable and trustworthy source of information coming from a recognized entity once it has been approved by the government. The FSR for the project was developed by a qualified designing institute.

CTI assessment team compared the values stated in the PD & MR with values determined in the FSR /7/, and was able to confirm that the input values are applied correctly in the PD & MR.

Item	Unit	Value	Source
Capacity	MW	50	FSR
Total Investment	10 ⁴ USD	9,806.15	FSR
Static investment	10 ⁴ USD	9,594.15	FSR
Tariff (excluding tax)	USD/kWh	0.17	FSR
First year attenuation rate	%	3.00%	FSR
Attenuation rate of each year since then	%	0.50%	FSR
Scheduled repayment period	year	12	FSR
Long-term loan interest rate	%	7.07%	FSR
Annual net Electricity Supplied	MWh	92,192	FSR
Operation Life	Year	20	FSR
Interest during construction period	10 ⁴ USD	212	FSR

Item	Unit	Value	Source
Personnel quota standard	Person	11	FSR
Average Labor Wage	10 ⁴ USD/year	1.93	FSR
Comprehensive welfare expenses	-	70%	FSR
Insurance rate of fixed assets	-	0.25%	FSR
Material cost	USD/kW	0.99	FSR
Other cost	USD/kW	5.68	FSR
Surplus accumulation fund percentage	-	10%	FSR
Working capital	USD/kW	6.4	FSR
Capital ratio	-	30%	FSR
Liquidity capital ratio	-	30%	FSR
Depreciation period	year	20	FSR
Residual rate	-	5%	FSR
Rate of Repair Fee during the first two years	-	0%	FSR
Rate of Repair Fee from 3 rd year to 5 th year	-	0.60%	FSR
Rate of Repair Fee from the 6 th year to the 20 th year	-	1.10%	FSR

Sub-step 2d: Sensitivity analysis (only applicable to Options II and III)

Sensitivity Analysis						
Items	-10.00%	-5.00%	0.00%	5.00%	10.00%	Critical variation
Total static investment	15.51%	13.15%	11.11%	9.34%	7.79%	14.89%
Annual operation cost	11.87%	11.49%	11.11%	10.73%	10.34%	100.90%
Tariff (Excl. VAT)	7.18%	9.10%	11.11%	13.21%	15.39%	16.16%-
Annual net electricity supplied	7.18%	9.10%	11.11%	13.21%	15.39%	16.16%-

Total static Investment

If the total static investment decreased by 14.89%, the EIRR would reach benchmark. By checking the project final account approved by Bangladesh Investment Development Authority/21/, the assessment team confirmed that the investment cost of the project activity (11,000 10⁴USD) is 12.17% higher than the estimated total static investment (9,806.15 10⁴USD). Therefore, it's unlikely that the total static investment decreased by 14.89%.

Annual operation cost

The Project EIRR would reach the benchmark when the annual operation cost decreases by 100.90% which means the operation cost will be negative value to make the EIRR reaching benchmark. It is obviously impossible.

Annual net electricity supplied and electricity tariff (Electricity revenue)

The Project EIRR would reach the benchmark when the electricity sale revenue increased by 16.16%. The electricity supplied or plant load factor from the FSR is determined by a third-party design institute and approved by the government. According to the FSR, the data of solar resource used to estimate the electricity output from the project was determined based on historical meteorological data and one year on-site solar resources measurement data provided by local meteorological station, which follows the assessment method for solar energy resources. The data was processed in professional software to calculate the annual power generation. Hence the annual power generation represents a long-term power supply throughout the lifetime of the project and the yearly-variations have already been taken into account. Therefore, the average annual electricity output will not change substantially in subsequent years. Further, in the verification process, the assessment team verified the net electricity supplied to the grid in the monitoring period 04/11/2020 - 03/11/2027, and confirmed the reported value during the monitoring period is only 2.08% lower than estimation in the FSR. Hence, the net electricity supplied is unlikely to increased by 16.16%.

The tariff fixed in the power purchase agreement of the project is 0.17 USD/kWh (excluding tax) same as the value estimated in the FSR. Therefore, it is unlikely that the electricity revenue would increase by 16.16%.

Step3: Barriers analysis

Barrier analysis was not applied for the project activity.

Step 4: Common practice analysis

The common practice analysis was addressed according to CDM Tool 01 “Tool for the demonstration and assessment of additionality” (Version 07.0) and Tool 24 “Common practice” (Version 03.1).

The assessment team defined the criteria for identifying similar projects, which are:

- Non - CDM&VCS photovoltaic power projects which have been officially approved by Bangladesh;
- Photovoltaic power projects with an output capacity between 25-75MW ;
- Photovoltaic power projects put into commercial operation before 04/11/2020.

The assessment team considered that identifying PV projects that began commercial operation prior to the time this project began operation ensures maximum identification, and therefore the assessment team confirmed that is reasonable. Based on the above criteria, the assessment team believes that the identification results in the PD & MR were verified by cross-checking the Annual Report of Bangladesh Power Development Board/30/ and no similar projects were found.

Therefore, the assessment team can conclude that the project activity is not common practice in Bangladesh.

3.4.6 Quantification of GHG Emission Reductions and Removals

The assessment team checked the baseline, project and leakage calculation, and confirmed that the evaluation of baseline, project and leakage is following the approved methodology ACM0002 (Version 20.0), and formula used to calculate is correct. The detail analysis is as below:

Baseline Emission:

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid- connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emission in year y (tCO₂e/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”

The project activity is a Greenfield solar power plant, then:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$EF_{grid,CM,y}$ calculation

The combined margin CO₂ emission factor, $EF_{grid,CM,y}$, is determined ex-ante following the “Tool to calculate the emission factor for an electricity system” (Version 07.0) and the official data published by the host country.

STEP 1. Identify the relevant electricity systems

Power Grid Company of Bangladesh is the sole grid operating agency and wheels the entire grid connected power. It is a government owned company operating in Bangladesh and a subsidiary of

Bangladesh Power Development Board. Therefore, information of Step 1 in the joint PD & MR is correct.

STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional).

For this project, only grid power plants are included in the calculation (option 1), which is reasonable. Off-grid generation is not important for the country because the main electricity consumption in Bangladesh is supported by the national grid.

STEP 3. Select a method to determine the operating margin (OM).

In Bangladesh, low-cost/must-run (LC/MR) power plants include hydro and renewable power plants. The quantity of electricity generated by these power plants constitutes less than 50% of total grid generation in average of the five most recent year/30/.

STEP 4. Calculate the operating margin emission factor according to the selected method.

PP selects Option B for $EF_{grid,OM}$ calculation based on the available data, and the assessment team determined that the $EF_{grid,OM}$ calculation is correct through the documentation/30//32/ and the ER calculation table/2/.

The assessment team has cross-checked all data with its data source: The quantity of electricity generated and delivered to the national grid, the type of fossil fuel consumed in electricity generation, and the energy conversion efficiency all have been checked with BPDB Annual Audit Report Year 2017-2020/30/. The CO₂ Emission Factor of fossil fuel follows IPCC default values as specified in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories/32/.

The Operating Margin CO₂ emission factor in year y ($EF_{grid,OM,y}$) is calculated using a 3-year data vintage as 0.5992 tCO_{2e}/MWh.

STEP 5. Calculate the build margin (BM) emission factor

PP has chosen option 1 to estimate the Build Margin (BM) ex-ante for the entire crediting period of the project activity.

Through the Annual Report of Bangladesh Power Development Board, the assessment team confirmed 2019-2020 as the most recent historical years for which generation data are available, and therefore considered it reasonable to use 2019-2020 data for $EF_{grid,BM}$ calculation.

The assessment team has cross-checked all data with its data source: The quantity of electricity generated and delivered to the national grid, the type of fossil fuel consumed in electricity generation, and the energy conversion efficiency all have been checked with BPDB Annual Audit Report Year 2017-2020/30/. The CO₂ Emission Factor of fossil fuel follows IPCC default values as specified in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories/32/.

The Build Margin CO₂ emission factor in year y ($EF_{grid,BM,y}$) is calculated as 0.5300 tCO₂/MWh.

STEP 6. Calculate the combined margin (CM) emissions factor.

The weighting of EF_{OM} and EF_{BM} for solar power generation project has been adopted as per “Tool to calculate the Emission Factor for an electricity system” (version 07.0)/44/ i.e. $W_{OM} = 0.75$ and $W_{BM} = 0.25$ for the first crediting period and for subsequent crediting periods. The calculation in the joint PD & MR uses the correct value of W_{OM} and W_{BM} . The Combined Margin emissions factor ($EF_{grid,CM,y}$) is $0.5992 \times 0.75 + 0.5300 \times 0.25 = 0.5818$ (tCO₂e/MWh)

$EF_{grid,CM,y} = 0.5818$ tCO₂e/MWh, which is fixed ex-ante for the first crediting period. $EG_{PJ,y}$, annual net power generation data from the FSR.

Project Emission:

As per the approved ACM0002 (Version 20.0), the project activity is a solar power farm, belongs to renewable energy activity, $PE_y = 0$.

Leakage Emission:

Leakage emission is not considered following ACM0002 (Version 20.0).

Emission reduction:

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e)

BE_y = Baseline Emissions in year y (tCO₂e)

PE_y = Project emissions in year y (tCO₂e)

And the baseline emissions (BE_y in tCO₂e) are the product of the baseline emissions factor ($EF_{grid,CM,y}$ in tCO₂/MWh) times the net electricity supplied by the project activity to the grid ($EG_{facility}$ in MWh). The calculation formula is as follows:

$$BE_y = EG_{facility,y} \times EF_{grid,CM,y}$$

The summary of ex ante estimates of emission reductions is shown as follows:

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 1	56,076	0	0	56,076

Year 2	55,818	0	0	55,818
Year 3	55,560	0	0	55,560
Year 4	55,302	0	0	55,302
Year 5	55,044	0	0	55,044
Year 6	54,786	0	0	54,786
Year 7	54,527	0	0	54,527
Total	387,113	0	0	387,113

3.4.7 Methodology Deviations

The validation process has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (Version 20.0) /43/. There was not any methodology deviation applied to the project activity.

3.4.8 Monitoring Plan

Assessment team checked the monitoring practice onsite and also checked the guideline of respective State electricity regulatory commission. The detail analysis is as below.

The parameters determined ex-ante

Data and Parameters	Unit	Ex-ante Determined Value
Operating margin (OM)	tCO _{2e} /MWh	0.5992
Build margin (BM)	tCO _{2e} /MWh	0.5300
Weighting of operating margin emissions factor (W_{OM})	-	0.75
Weighting of build margin emissions factor (W_{BM})	-	0.25
Combined emission factor (CM)	tCO _{2e} /MWh	0.5818

The parameters monitored ex-post:

Paramet	Description	Measurement method and QA/QC	Assessment
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ers		procedures	conclusion
EG _{facility,y}	Quantity of net electricity generation supplied by the project to the Grid in year y.	The parameter would be calculated as: $EG_{\text{facility},y} = EG_{\text{export},y} - EG_{\text{import},y}$ Monitored by the bi-directional main meter and backup meter with accuracy level no lower than 0.5. The readings of the electricity meters will be continuously measured and at least monthly recorded. The data will be cross checked with sales receipts.	Consistent with methodology

Monitoring Arrangement:

To ensure the implementation of project monitoring, PP installed two monitoring meters (M1 and M2) with no lower than 0.5 accuracy on the 132kV unit transformer side of the project site. The assessment team confirmed that the meter locations were as described in the joint PD & MR through remote site inspection by viewing meters' installation locations.

Organizational structure and responsibilities:

The responsibilities and authorities of project management, data handling and recording, measurement methods and QA/QC procedure have been systematically established and formalized and the same was verified during the online visit.

Data Measurement & Data collection and archiving:

As per the Joint PD & MR, the meter reading records of the master meter M1 will be used for emission reduction calculation and the invoices issued by the grid company will be used for cross-checking. The VVB will use the same source for verification of emission reductions. The meter readings are recorded by the project operator and invoiced by the national power company on a specific day of the month, as agreed in the PAA. When the main meter is out of service, the data monitored by the backup meter is used. The agency is experienced in the monitoring system and is managing O&M of numerous other Solar rooftop projects. The assessment team therefore is of the opinion that the PP is capable of implementing the monitoring plan in the context of the project activity.

QA/QC Procedures:

Energy meters with an accuracy of no lower than 0.5 are installed at the project site. Calibration of all meters is performed annually by an appropriately qualified third party in accordance with industry standards. Export and import power will be measured continuously using the above-mentioned meters installed on the 132kv side of the unit transformer. The meter readings will be read by trained staff and

confirmed by the project manager and then saved. Invoices from the grid company can be used to cross-check meter readings for the project activity.

It is reported that the data will be kept for 2 years following the end of the crediting period.

Personnel training:

In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan. The CTI validation & verification team confirmed the above by reviewing the monitoring manual set by PP/23/ and confirmed the implementation of the training through the training records.

3.5 Non-Permanence Risk Analysis

Not applicable for the project activity.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a solar power grid-connected plant and locates a 170-acre plot of land approximately 7km South from the town center of Mymensingh City, Bangladesh. The geographical coordinates are verified by GPS system as 24°42'13.39"N (24.70372°) and 90°27'49.81"E (90.46384°). The project involves the installation and operation of 60.251 MWp DC (50.05 MW at AC level) solar power plant equipped with 88,356 photovoltaic (PV) solar modules in type of LR4-72HDB-435M and 50,736 PV solar modules in type of LR4-72HDB-430M with 20 years lifetime, manufactured by LONGI Solar Technology Co., Ltd. It was confirmed by checking the operation logs/24/, Construction Complete Report/25/ and Commercial Operation Certificate/26/ that the project began to construct on 22/09/2019 and started commercial operation on 04/11/2020. The actual implementation of the project during this verification period (04/11/2020 to 31/08/2022) was verified in terms of nameplates of PV solar modules /15/. The details of the PV solar modules with respect to their installation and capacity have been verified to be consistent with description indicated in the joint PD & MR.

By reviewing the Power purchase agreement (PPA) signed between HDFC SinPower Limited and Bangladesh Power Development Board (BPDB)/10/ and diagram of power connection system/16/, it can be confirmed that the DC power produced by PV modules is fed into grid-connected inverter. In the grid-connected inverter, the DC current is converted into AC current, flowing to 33kV step-up transformer. After that, 33kV voltage comes out and is going to be put into evacuation switch yard where the power will be step up into 132 kV and transported to nearby Kewatkhali substation. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment.

By checking the operation logs/24/ and via remote site inspection, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period. All the monitoring system in operation period is consistent with the description in monitoring plan.

On-site training for the related procedures including, recording and reporting was verified to be in place/35/ and their implementation was confirmed by interview with the key operators and observing the operation.

Via remote site inspection and document reviews i.e. checking Operation logs /24/ and evidences regarding monitoring activities/23/,/27/-/29/, the assessment team confirmed that:

- the project implementation is in accordance with the registered PD;
- there is no other proposed or actual change to the registered PD during this monitoring period;
- all required equipment and procedures are available and implemented in an appropriate manner;
- the project is completely operational and all necessary monitoring instruments are installed;
- all required instruments including standby and operating procedures for the same have been implemented in an appropriate manner;
- neither mistakes nor malfunction on monitoring meters have been observed during this monitoring period.
- the control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment;
- no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

In conclusion, as part of the remote audit, CTI confirms that the project implementation is in accordance with the project description contained in the joint PD & MR. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the monitoring plan contained in the joint PD & MR is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (Version 20.0).

Compliance of monitoring with the monitoring plan

The monitoring has been carried out in accordance with the monitoring plan contained in the joint PD & MR/1/, CTI confirms that all parameters stated in the monitoring plan are monitored and reported appropriately. All parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (Version 20.0) and the management system were assessed during the interview. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e.

from data generation, recording, aggregation, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

Parameters fixed ex ante

The parameter $EF_{grid,CM,y}$ is the combined margin emission factor of the grid, which was determined ex-ante at the validation stage and was fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the joint PD & MR.

Parameters monitored

CTI conducted document review and performed remote site inspection with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and electricity receipts;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the monitoring plan, the parameters to be monitored is as below:

$EG_{facility,y}$	Quantity of net electricity generation supplied by the project to the Grid in year y.
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Monitored parameters $EG_{facility,y}$

The parameter is the net amount of electricity delivered to the grid by the project, which is the difference between the electricity exported and the electricity imported. The electricity exported and the electricity imported are measured by main meter M1 and backup meter M2. Both meters are bi-directional and located at the 132kV side of the unit transformer within the project site. The meter readings of electricity exports and imports were continuously measured, daily read and monthly aggregated in line with the applied methodology and monitoring plan.

The assessment team had online checked the location of the meters against the diagram of power connection system /16/ and found them to be consistent and installed in accordance with the monitoring plan. All the monitoring facilities and system have been verified by CTI during remote audit.

By checking operation log /24/ and via remote site inspection, it is confirmed that both meters were working well during this monitoring period. Via the online interview and by checking the daily operation records/24/, daily and monthly reading records and electricity receipts/27/, it is confirmed by the assessment team that the cut-off time of electricity transaction is at 24:00 of the last day of each month/27/. By interviewing with PP, the assessment team found the net electricity

supplied to the grid (electricity exported minus to electricity imported) was used to settle accounts between PP and grid company, and was confirmed by checking the PPA. Hence, the electricity sales receipts/28/ issued by the grid company have been used to cross-check the records of net electricity supplied to the grid.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the monitoring plan in the joint PD & MR and the methodology ACM0002 (Version 20.0) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

Meters	Type	SN	Accuracy class	Calibration Date	Validity	Calibration Entity
Main meter (M1)	ZMD402CT	38992612	0.2S	30/10/2020 13/10/2021	29/10/2021 12/10/2022	Bangladesh Power Development Board
Backup meter (M2)	ZMD402CT	38992613	0.2S	30/10/2020 13/10/2021	29/10/2021 12/10/2022	

The calibration of electricity meters is performed by the grid company. Calibration records /29/ have been verified by the verification team. The actual accuracy of meters M1 and M2 is verified to be 0.2S, which is consistent with the requirement in the monitoring plan (accuracy degree is not lower than 0.5) contained in the joint PD & MR. The calibration frequency of meters is annual, which is consistent with the monitoring plan (The metering equipment will be properly calibrated and checked annually). Through the remote auditing, by checking the power connection diagram and calibration reports, CTI confirms that:

- All meters have been installed in accordance with the monitoring plan;
- The meters' accuracy is in line with the requirement of the monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the monitoring plan. The calibrations are verified to be valid for the whole reporting period.

Data management and control

HDFC SinPower Limited is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/23/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff and check records /24//27/, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan contained in the joint PD & MR and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

Participation or any rejection under any other GHG programs, any other form of environmental credit or emissions trading program since validation or previous verification

The verification team has also checked GHGs programs and forms of credit such as the clean development mechanism (CDM), Gold Standard, EU ETS, IREC, and based on the interview with the PP, during this monitoring period, it is confirmed that except VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification;
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification.

SD contributions resulted by implementation of the project activity

Via the remote site inspection and interviews with the PP, local villagers and official from local environment bureau, and by checking the evidences provided by the PP, the assessment team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- SDG 7, Indicator 7.2 Renewable energy supplied to the power grid.

The project provides clean and renewable energy through making use of solar power, displacing fossil fuel power in the grid (BPDB). During this monitoring period, 171,859.680 MWh of net electricity generated by solar power was delivered to the BPDB. It is verified by checking the daily and monthly reading records/27/ and via online inspection.

- SDG 8, Indicator 8.5 Numbers of job opportunities provided by implementation of the project.

For 2019-2020, senior management salaries in Bangladesh range from \$400-\$500 per month/31/. Through interviews with PPs, employees and local villagers, by checking staff list, salary certificate /5/, it was confirmed that the average monthly salary of the program's employees is \$633, which far exceeds the salaries of senior management. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable.

- SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed.

By remote site inspection, checking the emission reduction calculation sheet/2/ and the daily and monthly reading records/27/, 99,987 tCO₂e GHG emission reductions are generated by the project activity during this monitoring period.

This is the first verification of the project and also the first application for SD contribution, so the cumulative contribution in “Contributions Over Project Lifetime” in the joint PD & MR is consistent with the current monitoring period.

In conclusion, the assessment team was able to confirm that the project implementation is in accordance with the project description contained in the joint PD & MR. All required equipment and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed. Neither mistakes nor malfunction on meters have been observed during this monitoring period.

4.2 Accuracy of GHG Emission Reduction and Removal Calculations

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity supplied to the grid by the project during year y ($EG_{\text{facility},y}$) by the validated ex-ante fixed grid emission factor ($EF_{\text{grid},CM,y}$).

$$BE_y = EG_{\text{facility},y} \times EF_{\text{grid},CM,y}$$

Combined margin CO₂ emission factor for grid connected power generation ($EF_{\text{grid},CM,y}$)

$EF_{\text{grid},CM,y}$ is the grid emission factor of the which has been verified ex-ante in the validation stage in the joint PD & MR as 0.5818 tCO₂e/MWh.

The net electricity supplied to the grid by the project ($EG_{\text{facility},y}$)

As per the joint PD & MR, $EG_{\text{facility},y}$ is determined as the following formula:

$$EG_{\text{facility},y} = EG_{\text{export},y} - EG_{\text{import},y}$$

The assessment team has checked the daily and monthly reading records, electricity receipts, and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The reading records and calculation are summarized as follow:

Period	EG _{export,y}	EG _{import,y}	EG _{facility,y}		
	Values from Main Meter Reading Record	Values from Main Meter Reading Record	Calculated based on the Meter Readings	Values from Electricity Sales Receipts	Conservative values
	A	B	C=A-B	D	C=MIN (A, B)
	MWh	MWh	MWh	MWh	MWh
04/11/2020~30/11/2020	8,309.568	38.592	8,270.976	8,270.976	8,270.976
01/12/2020~31/12/2020	6,198.144	45.600	6,152.544	6,152.544	6,152.544
Subtotal in 2020	14,507.712	84.192	14,423.520	14,423.520	14,423.520
01/01/2021~31/01/2021	4,629.792	44.160	4,585.632	4,585.632	4,585.632
01/02/2021~28/02/2021	7,374.336	40.368	7,333.968	7,333.968	7,333.968
01/03/2021~31/03/2021	8,721.168	39.984	8,681.184	8,681.184	8,681.184
01/04/2021~30/04/2021	10,029.312	35.808	9,993.504	9,993.504	9,993.504
01/05/2021~31/05/2021	9,117.840	36.960	9,080.880	9,080.880	9,080.880
01/06/2021~30/06/2021	7,374.480	33.456	7,341.024	7,341.024	7,341.024
01/07/2021~31/07/2021	7,604.016	35.328	7,568.688	7,568.688	7,568.688
01/01/2021~31/08/2021	6,913.392	36.000	6,877.392	6,877.392	6,877.392
01/09/2021~30/09/2021	9,201.600	35.904	9,165.696	9,165.696	9,165.696
01/10/2021~31/10/2021	8,966.736	41.568	8,925.168	8,925.168	8,925.168
01/11/2021~30/11/2021	8,721.216	44.976	8,676.240	8,676.240	8,676.240
01/12/2021~31/12/2021	7,179.216	48.768	7,130.448	7,130.448	7,130.448
Subtotal in 2021	95,833.104	473.280	95,359.824	95,359.824	95,359.824

Period	EG _{export,y}	EG _{import,y}	EG _{facility,y}		
	Values from Main Meter Reading Record	Values from Main Meter Reading Record	Calculated based on the Meter Readings	Values from Electricity Sales Receipts	Conservative values
	A	B	C=A-B	D	C=MIN (A, B)
	MWh	MWh	MWh	MWh	MWh
01/01/2022~31/01/2022	5,651.232	41.616	5,609.616	5,609.616	5,609.616
01/02/2022~28/02/2022	7,725.504	40.848	7,684.656	7,684.656	7,684.656
01/03/2022~31/03/2022	9,353.232	40.224	9,313.008	9,313.008	9,313.008
01/04/2022~30/04/2022	7,827.552	35.040	7,792.512	7,792.512	7,792.512
01/05/2022~31/05/2022	8,374.992	38.352	8,336.640	8,336.640	8,336.640
01/06/2022~30/06/2022	5,650.992	37.104	5,613.888	5,613.888	5,613.888
01/07/2022~31/07/2022	8,741.424	38.736	8,702.688	8,702.688	8,702.688
01/08/2022~31/08/2022	9,063.120	39.792	9,023.328	9,023.328	9,023.328
Subtotal in 2022	62,388.048	311.712	62,076.336	62,076.336	62,076.336
Total	172,728.864	869.184	171,859.680	171,859.680	171,859.680

Note: The invoices issued by the project company were based on the net electricity supply, and were used to cross-checking the net electricity supply..

Emission reductions

The emission reductions achieved by the project activity during this monitoring period is calculated as:

$$ER_y = BE_y - PE_y - L_y = 99,987 tCO_2e$$

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates ex-ante

The emission reductions claimed are 99,987 tCO₂e in this monitoring period. Compared with expected emission reductions calculated ex-ante in the jointly PD & MR, the reported emission reductions in this monitoring period are 2.08% less than that expected, which is considered to be in the reasonable variation range.

Verified GHG emission reductions and removals from 04/11/2020 to 31/08/2022:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
04/11/2020-31/12/2020	8,391	0	0	8,391
01/01/2021-31/12/2021	55,480	0	0	5,5480
01/01/2022-31/08/2022	36,116	0	0	36,116
Total	99,987	0	0	99,987

CTI is able to confirm that the actual emission reductions reported in this monitoring period are reasonable and appropriate. CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent. And CTI is able to confirm that GHG emission reductions and removals have been quantified correctly in accordance with the monitoring plan and applied methodology for this verification period.

4.3 Quality of Evidence to Determine GHG Emission Reductions and Removals

The assessment of each parameter related to calculate the GHG emission reductions is justified in Sections 4.2 above. Via the remote site inspection and desk review, the assessment team verified the supporting evidence provided by the project proponent as listed in Appendix 2 used to determine the GHG emission reductions.

For each reported data, the evidence is provided and verified as sufficient and quality is appropriate. Also, the cross-checks have been performed on the reported data with different source of evidence. The information flow from data generation and aggregation, to recording, calculation and final transposition

into the monitoring report has been assessed by VVB in Section 4.1. Moreover, the calibration is verified as stated in Section 4.1 via online inspection with evidences to confirm the meters' location, accuracy and calibration frequency are in line with requirement of the monitoring plan.

Therefore, it is concluded that the evidence provided are verified as sufficient and quality is appropriate and thus the evidence can be used to determine the GHG emission reductions and removals for this monitoring period.

Based on the above, the assessment team confirms the sufficiency and appropriateness of the quality of evidence provided by the PP to determine the GHG reductions and further deems them to be acceptable.

5 VALIDATION AND VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the joint validation and verification of the project activity "50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh" in Bangladesh (VCS Project ID: 3545) for the monitoring period 04/11/2020 to 31/08/2022 within the first crediting period 04/11/2020 - 03/11/2027.

The review of the project design documentation and the subsequent follow-up interviews have provided CTI with sufficient evidence to determine the fulfilment of stated criteria.

In the course of the validation and verification, 6 Clarification Requests (CLs), 4 Corrective Action Requests (CARs). No Forwarded Action Request (FAR) was raised.

The review of the project design documentation and additional documents related to baseline and monitoring methodology and subsequent background investigation have provided the CTI with sufficient evidence to validate the fulfilment of the stated criteria. In detail the conclusions can be summarized as follows:

- the baseline scenario is correctly defined as per the applied methodology and relate tools;
- the project is additional;
- all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature;
- a reasonable level of assurance has been applied;
- the monitoring plan in the validated PD is as per the applied methodology;
- the project has been implemented and operated as per the joint PD & MR;
- the project complies with the validation criteria for projects set out in VCS Standard;

- The calculation of the ex-ante emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of 387,113 tCO₂e are most likely to be achieved within the first 7-year renewable crediting period started from 04/11/2020.

It is CTI's responsibility to provide an independent validation statement on the Project's compliance with the VCS Program requirements as set out in the VCS Program Rules. The conclusions of this report show that the project, as it was described in the project description contained in the joint PD & MR, is in line with all criteria applicable for the validation against the VCS standard without any qualifications or limitations.

The verification is based on the baseline and monitoring methodology ACM0002 (Version 20.0), and the joint PD & MR (Version 02 dated 27/10/2022). The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the joint PD & MR. Our verification approach was based on the requirements as defined under the applicable VCS Standard and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the monitoring plan in joint PD & MR is as per the applied methodology;
- the monitoring complies with the monitoring plan in the joint PD & MR;
- the joint PD & MR and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCU's hereby verified and certified. Request for issuance of VCU's shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh" for the monitoring period 04/11/2020 to 31/08/2022 are fairly stated in the joint PD & MR (Version 02 dated 27/10/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (Version 20.0) and the monitoring plan contained in the joint PD & MR (Version 02 dated 27/10/2022).

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Verification period: From 04/11/2020 to 31/08/2022

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
04/11/2020-31/12/2020	8,391	0	0	8,391
01/01/2021-31/12/2021	55,480	0	0	5,5480
01/01/2022-31/08/2022	36,116	0	0	36,116
Total	99,987	0	0	99,987

Wu Lin

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29-Oct-2022

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Technical Reviewer
29-Oct-2022

APPENDIX 1: ABBREVIATIONS

Abbreviations	Full texts
AFOLU	Agriculture, Forestry, and Other Land Use
BM	Build Margin
CDM	Clean Development Mechanism
CM	Combined Margin
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
GHG	Greenhouse gas(es)
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETS	Emission Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse gas(es)
GS	Gold Standard
IPCC	Intergovernmental Panel on Climate Change
MP	Monitoring Plan
OM	Operating Margin
PD	Project Description
PP	Project Proponent
PPA	Power Purchase Agreement
PV	Photovoltaic
QA/QC	Quality Assurance / Quality Control

Abbreviations	Full texts
SDG	Sustainable Development Goal
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit
VVB	Validation/Verification Body

APPENDIX2: REFERENCE LIST

/1/	VCS joint PD & MR, Version 01 dated 15/07/2022, Version 02 dated 27/10/2022
/2/	Emission reduction spreadsheet
/3/	IRR spreadsheet
/4/	Prior consideration Email and VCS Form as evidence of VCU for the project activity dated on 25/05/2019
/5/	HDFC SinPower Limited: Staff list and salary certificate
/6/	Environmental Impact Assessment (EIA) prepared by EQMS Consulting Limited, 03/2017 EIA approval by Depantrnemt of Environnnemt, Government of the People's Republic of Bangladesh, dated 20/04/2017
/7/	Feasibility Study Report (FSR) prepared by Fujian Yongfu Electric Power Design Co., Ltd, 09/2019
/8/	Implementation agreement by and between the government of the People's republic of Bangladesh and PGCB and HDFC Sinpower Limited, 19/10/2016
/9/	Approval of the project, China NDRC, 12/03/2020
/10/	Power Purchase Agreement signed with Bangladesh Power Development Board
/11/	Grid Access Agreement issued by Bangladesh Power Development Board, dated on 04/11/2020
/12/	Business license of HDFC SinPower Limited
/13/	Business license of China International Engineering Consulting Corporation (Hong Kong) Limited
/14/	Purchase contract and technical specifications supportive of photovoltaic modules
/15/	Nameplates of the photovoltaic modules
/16/	Construction layout diagram of photovoltaic power plant
/17/	Statement of no-double counting by project proponent, dated 20/09/2022
/18/	The website of Asian Development Bank https://www.adb.org/countries/bangladesh/economy
/19/	Fisher Equation, https://en.wikipedia.org/wiki/Fisher_equation

/20/	Guidelines on countries (regions) for foreign investment cooperation - Bangladesh
/21/	Project final account of 50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh approved by Bangladesh Investment Development Authority
/22/	Laws and regulations: Renewable Energy policy (Renewable Energy Policy of Bangladesh -2008)
/23/	HDFC SinPower Limited: VCS monitoring manual and management procedure
/24/	HDFC SinPower Limited: Operation logs
/25/	Construction Complete Report
/26/	Commercial Operation Certificate issued by the power grid company on 21/01/2021
/27/	HDFC SinPower Limited: Daily and monthly reading records of 50 MW (AC) Solar Park at Sutiakhali, Gouripur, Mymensingh from 04/11/2020 to 31/08/2022
/28/	HDFC SinPower Limited: Invoices for net electricity generation supplied by the project plant to the grid for 04/11/2020 to 31/08/2022
/29/	Bangladesh Power Development Board: Calibration certificates for meters M1 and M2 covering this monitoring period 04/11/2020 to 31/08/2022
/30/	Annual Report of Bangladesh Power Development Board 2015-2020
/31/	Guidelines on countries (regions) for foreign investment cooperation - Bangladesh, 2021 Edition
/32/	IPCC 2006, Guidelines for National Greenhouse Gas Inventories
/33/	Stakeholder Questionnaire
/34/	Meeting minutes for group discussions
/35/	HDFC SinPower Limited: Records of training for on-site staff.
/36/	VCS Standard, version 4.3
/37/	VCS Programme Guide, version 4.2
/38/	VCS Validation and Verification Manual, version 3.2
/39/	VCS Joint project description & monitoring report template, Version 4.1
/40/	CDM Validation and Verification Standard for project activities, version 03.0

/41/	CDM Project Standard for project activities, version 03.0
/42/	CDM project cycle procedure for project activities, version 03.0
/43/	Approved methodology ACM0002 (version 20.0), dated 28/11/2019
/44/	Tool to calculate the emission factor for an electricity system, version 07.0
/45/	Applicability of sectoral scopes, version 01.0
/46/	UN 17SDGs: https://www.un.org/sustainabledevelopment/sustainable-development-goals/ UNFCCC: http://cdm.unfccc.int VCS: http://verra.org GS: https://www.goldstandard.org

APPENDIX3: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Lin Wu

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.1: Thermal energy generation
	TA 1.2: Energy generation from renewable energy sources
SS 2: Energy distribution	TA 2.1: Electricity distribution
SS 3: Energy demand	TA 3.1: Energy demand
SS 4: Manufacturing industries	TA 4.1: Cement and lime production
SS 5: Chemical industry	TA 5.1: Chemical industry
	TA 5.2: Caprolactam, nitric and adipic acid
SS 10: Fugitive emissions from fuels (solid, oil and gas)	TA 10.1: Fugitive emissions from oil and gas
SS 11: Fugitive emissions from production and consumption of halocarbons and sulphur hexafluoride	TA 11.1: Emissions of fluorinated gases
	TA 11.2: Refrigerant gas production

SS 12: Solvents use	TA 12.1: Chemical industry
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater
	TA 13.2: Manure

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:



Lu ZHOU

General Manager

Shenzhen, 01/01/2021

Ms. Zhang Wenting

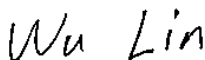
Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	-	-	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:



Wu LIN

Technical Competent Manager

Shenzhen, 01/03/2022

Mr. Nasif Shams

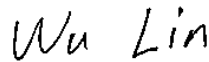
Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	-	-	-	-	-	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:



Wu LIN

Technical Competent Manager

Shenzhen, 01/07/2022

Ms. Lin Shunrong

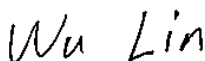
Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:



Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX4: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Validation & Verification Team Assessment
CAR01	The format of the catalog is not consistent with the requirements in the template.	The format of PD&MR has been updated as per the PD&MR template.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CAR01 is closed.
CAR02	The project size in Section 1.3 and 1.10 is not consistent with the requirement in the Standard V4.3.	The project size has been corrected to be 'project' as per the requirement of VCS Standard V4.3.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CAR02 is closed.
CAR03	The calculation method of "Contributions Over Project Lifetime" in Section 1.17.2 is not consistent with the template.	The description in the Table 1 of Section 1.17.2 has been corrected to the cumulative impact calculated by summing the current project contributions as per the PD&MR template.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CAR03 is closed.
CAR04	The end day of this monitoring period in Section 6.5 is not consistent with the ER table	This monitoring period is from 04/11/2020 to 31/08/2020 and the end day of the monitoring period has been corrected to be consistent with the ER spreadsheet.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CAR04 is closed.

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Validation & Verification Team Assessment
CL 01	Section 1.11 lacks the detail information of the main equipment.	The detailed information has been complemented for the main equipment in the updated PD&MR.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CL01 is closed.
CL02	The capacity of the solar PV modules in Section 1.11 is inconsistent with the feasibility study. Please clarify the inconsistency and whether this inconsistency will affect the power generation and thus the additionality.	In the FSR, the designed capacity is 60.177MWp at DC level with the installation of 139,947 pieces of solar PV modules with single capacity of 430Wp. Actually, as per the requirements of on-site installation, the 88,356 pieces of modules with single capacity of 435Wp and 50,736 pieces of with single capacity of 430Wp were installed and constitute the total capacity of 60.251 MWp at DC level. On the one hand, it is considered that the designed total capacity is similar with the actually installed capacity. It is common in solar PV power generation industry that there is a slight difference between the designed installed capacity and the actual installed capacity. On the other hand, the actually installed capacity of inverters is consistent with the designed capacity in the FSR, i.e. 286 inverters with each capacity of 285 KVA (175kW), which ensures the same output capacity 50MW at AC level. Therefore, the power generation does not change substantially due to the slight difference of the total capacity and it does not affect additionality of the project.	OK. Based on the sectoral experience, the assessment team considered the change above is the common issues in the solar power field, and the slight change in the installation capacity will not affect greatly the power generation and financial values. It has been verified by CTI. Therefore, CL02 is closed.

CL03	The information of project location in Section 1.12 is not consistent with the FSR. The inconsistency is request to clarify.	It is a typo on the project location, which has been corrected as per the FSR.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CL03 is closed.
CL04	more financial parameters and data are requestee to be provided in Section 3.5.	The financial parameters and data have been complemented in Section 3.5 of updated version.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CL04 is closed.
CL05	The operating hours and tariffs in section 3.5 are inconsistent with the FSR. The inconsistency is request to clarify.	It is a typo. The operation hour is 1844h and tariff is 1.7 USD/kWh (excluding tax), which has been corrected to be consistent with the FSR.	OK. It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CL05 is closed.
CL06	Need to clarify in section 3.5 that changes in parameters do not affect the IRR.	The influence of financial parameter fluctuation on IRR has been added in the Section 3.5 of updated version.	OK It has been revised and verified by CTI by checking updated joint PD & MR. Therefore, CL06 is closed.

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Validation & Verification Team Assessment
NA	NA	NA	NA